A Novel Approach to Format Based Text Steganography

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ABSTRACT
Steganography gained importance in the past few years due to the increasing need for providing secrecy in an open environment like the internet. With almost anyone can observe the communicated data all around, steganography attempts to hide the very existence of the message and make communication undetectable. Steganography has many technical challenges such as high hiding capacity and imperceptibility. In this paper, we try to evolve an algorithm on text steganography using the combination of line shifting and word shifting methods with copy protection technique with high capacity of the cover object. Word shift embeds secret information in text by shifting words slightly. Line shift embeds secret information in text by shifting text lines slightly. Our method hides the secret information in binary form rather than the original character form by using the above methods. This method has a good hiding capacity because it hides more than one bit in each line of cover text. Also this method does not make any apparent changes in the original text.

Categories and Subject Descriptors
C.2.0 [**Computer-Communication Networks**]: General- *Security and protection*; E.3 [**Data**]: Data Encryption.

General Terms
Algorithms, Security.

Keywords
Steganography, Text steganography, Word shifting, Line shifting.

1. INTRODUCTION
Steganography is the art and science of writing hidden messages in such a way that no one, apart from the sender and intended recipient, suspects the existence of the message, a form of security through obscurity. The word "*steganography*" is of Greek origin and means "concealed writing" from the Greek words "*steganos*" meaning "covered or protected", and "*graphein*" meaning "to write". Steganography is a method for hidden exchange of information by hiding data in a cover media such as image or sound or text [6]. After the expansion of new technologies especially information technology and computer systems in recent decade, the issue of security of information has gained special significance. One of the main fields of information security is the concept of hidden exchange of information. For this purpose, various methods including cryptography, steganography, coding and so on have been used. Steganography is one of the methods in this field which have attracted more attention during the recent years.

In implementing steganography, the main objective of Steganography is to hide the information under a cover media so that the outsiders may not discover the information contained in the said media. This is the major distinction between Steganography and other methods of hidden exchange of information. For example, in cryptography method, people become aware of the existence of data by observing coded data, although they are unable to comprehend the data. However, in Steganography, nobody notices the existence of data in the resources. Most Steganography works have been performed on images, video clips, text, music and sounds. However text Steganography is the most difficult kind of Steganography; this is due to the lack of redundant information in a text file, while there is a lot of redundancy in a picture or a sound file, which can be used in Steganography. A digital text documents is written, saved and retrieved by a computer in exactly the same way as it appears before naked-eye in contrast to other file formats such as picture where the observed and saved contents are different and can serve the purpose of hiding secret information within its structure. The applications of information hiding in text vary from Natural Language Processing techniques to the HTML file format. The structure of text documents is identical with what we observe, while in other types of documents such as in a picture, the structure of document is different from what we observe. Therefore, in such documents, we can hide information by introducing changes in the structure of the document without making a notable change in the concerned output. Text Steganography is one of the most difficult methods because a text file is not a proper media to hide data in it. Among the most important of these technologies, one can name the hiding information in electronic texts and documents (e-documents). The use of hiding information in text for web pages is another example. In addition to hidden exchange of information, Steganography is used in other areas such as copyright protection, preventing e-document forging and other applications. Today, the computer systems have facilitated hiding information in texts, and some technologies for hiding information in text have also developed.

ICCCS'11, February 12–14, 2011, Rourkela, Odisha, India.

This paper is organized as follows: Section 2 describes related background work. The proposed model is given in Section 3.The proposed algorithm for encoding and decoding is given in Section 4. Experimental details is presented in section 5. Section 6 presents the example of proposed algorithm. Performance analysis is given in Section 7. Finally, section 8 concludes the paper.

2. BACKGROUND WORK

Before giving a brief description of the related work on Text Steganography, it will be appropriate to discuss the frequent terms used in this context. The secret message to be hidden is referred to as embedded data and the innocuous text / audio / image used for embedding is called as cover. The resultant output object after embedding is referred to as stego-object. It is a priory for the sending and receiving ends to have agreed upon on a mutual key exchange protocol / mechanism. A few works have been done on hiding information in texts. Following is the list of different methods of the works carried out and reported so far. In general, the Text Steganography methods can be categorized into two groups:

1- Changing the text format.

2- Changing the meaning of the text.

The methods which are based on the changing the meaning of the text are limited. Some examples of these methods are as follows:

2.1.1 Syntactic method

By placing some punctuation marks such as full stop (.) and comma (,) in proper places, one can hide information in a text file. This method requires identifying proper places for putting punctuation marks. The amount of information to hide in this method is trivial [6, 8, 9].

2.1.2 Semantic method

This method uses the synonym of certain words thereby hiding information in the text. The synonym substitution may represent a single or multiple bit combination for the secret information. A major advantage of this method is the protection of information in case of retyping or using OCR programs. However, this method may alter the meaning of the text [3, 4, 8]. Table 1 shows list of words in the left column and there synonyms in the right column.

Table 1.

Big	Large
Small	Little
Chilly	Cool
Smart	Clever
Spaced	Stretched

2.1.3 Text abbreviation or Acronym

Another method for hiding information is the use of abbreviations. In this method, very little information can be hidden in the text. For example, only a few bits of information can be hidden in a file of several kilobytes [3, 7]. Table 2 shows the list of acronyms in the left column and its translation in the right column.

Table 2.

Acronym	Translation
2l8	Too late
ASAP	As Soon As Possible
C	See
CM	Call Me
F2F	Face to face

In this method words can be substituted with their abbreviations to represent the binary bit pattern of zero or one corresponding to the bits of secret information.

2.1.4 Change of Spelling

This method exploits the way words are spelled in British and American English for hiding secret information bits [3, 9].

Table 3 shows the list of some words which have different spellings in UK and US.

Table 3.

American Spelling	British Spelling
Favorite	Favourite
Criticize	Criticise
Fulfill	Fulfil
Center	centre

The methods which change the format of the text usually have a large capacity for hiding information. Some examples of these methods are as follows:

2.1.5 Line shifting method

In this method, the lines of the text are vertically shifted to some degree (for example, each line is shifted 1/300 inch up or down) and information are hidden by creating a unique shape of the text. This method is suitable for printed texts.

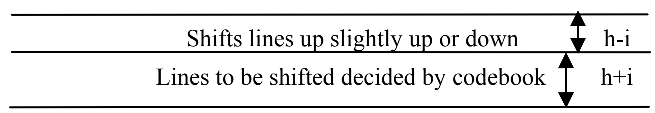


Figure 1. Line Shifting

However, in this method, the distances can be observed by using special instruments of distance assessment and necessary changes can be introduced to destroy the hidden information. Also if the text is retyped or if character recognition programs (OCR) are used, the hidden information would get destroyed. This method hides information by shifting the text lines to some degree to represent binary bits of secret information [3, 4, 5].

2.1.6 Word shifting method

In this method, by shifting words horizontally and by changing distance between words, information is hidden in the text. This method is acceptable for texts where the distance between words is varying. This method can be identified less, because change of distance between words to fill a line is quite common.

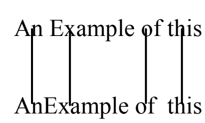


Figure 2. Word Shifting

But if somebody was aware of the algorithm of distances, he can compare the present text with the algorithm and extract the hidden information by using the difference. The text image can be also closely studied to identify the changed distances. Although this method is very time consuming, there is a high probability of finding information hidden in the text. Retyping of the text or using OCR programs destroys the hidden information [3, 4].

2.1.7 Open spaces or White Spaces

In open spaces method, hiding bits of secret information is done through adding extra white-spaces in the text. These white-spaces can be placed at the end of each line, at the end of each paragraph or between the words or sentences. This method can be implemented on any arbitrary text and does not raise attention of the reader. However, the volume of information hidden under this method is very little. Also, some text editor programs automatically delete extra white-spaces and thus destroy the hidden information [4].

T	**h**	**e**		**q**	**u**	**i**	**c**	**k**		**b**	**r**	**o**	**w**	**n**
f	**o**	**x**		**j**	**u**	**m**	**p**	**s**		**o**	**v**	**e**	**r**	
t	**h**	**e**		**l**	**a**	**z**	**y**		**d**	**o**	**g**	**.**		

Figure 3. Original Text

T	**h**	**e**		**q**	**u**	**i**	**c**	**k**		**b**	**r**	**o**	**w**	**n**	
f	**o**	**x**		**j**	**u**	**m**	**p**	**s**		**o**	**v**	**e**	**r**		
t	**h**	**e**		**l**	**a**	**z**	**y**		**d**	**o**	**g**	**.**			

Figure 4. Encoded Text

2.1.8 Feature Coding

In feature coding method, some of the features of the text are altered. For example, the end part of some characters such as h, d, b or so on, are elongated or shortened a little thereby hiding information in the text. In this method, a large volume of information can be hidden in the text without making the reader aware of the existence of such information in the text. This method hides the secret information bits by associating certain attributes to the text characters. By placing characters in a fixed shape, the information is lost. Retying the text or using OCR program destroys the hidden information. e.g., Steganograph*y* is the art of hiding secret information [4].

2.1.9 Steganography of Information in Specific Characters in Words

In this method, some specific characters from certain words are selected as hiding place for information. In the simplest form, for example, the first words of each paragraph are selected in a manner that by placing the first characters of these words side by side, the hidden information is extracted. This has been done by classic poets of Iran as well. This method requires strong mental power and takes a lot of time. It also requires special text and not all types of texts can be used in this method [7].

2.1.10 Creating Spam Texts

This method is based on the Internet. Based on the piece of information aimed for hiding, a number of spam emails are selected and some phrases appropriate to the concerned piece of information are chosen from among these letters, so that the concerned piece of information might be extracted from these phrases later. As 82 percent of emails in the United States are spam, this method does not attract too much attention [7].

2.1.11 HTML Tags

HTML Tags can be used in varying combination or as gaps and horizontal tabulation to represent a pattern of secret information bits [3].

For example, Stego key.

<img></img> -> 0

<img/> -> 1

Stego data:

<img src=g1.jpg></img>

<img src=g5.jpg></img>

Hidden Bits: 01110

2.1.12 XML Document

XML is a preferred way of data manipulation between web-based applications. The user defined tags are used to hide actual message or the placement of tags represents the corresponding secret information bits [3].

3. THE PROPOSED MODEL

We propose a new hybrid model using Special Character, Line Shifting and Word Shifting coding techniques of text steganography (see Figure 5).

3.1 Block Diagram of the Proposed Model

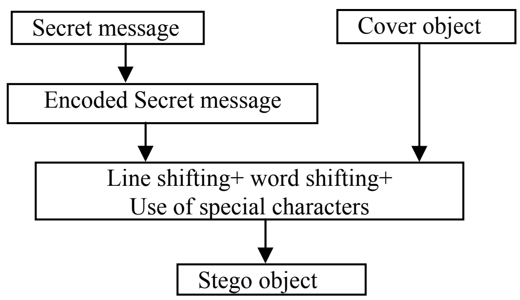


Figure 5. Broad level steps in Text Steganography

4. THE PROPOSED ALGORITHM

4.1 Proposed Algorithm for Encoding

Step 1: Count the no of bits in the secret message. If the no is odd then add a "0" at the left and go to step 2, otherwise after counting directly go to step 2.

Step 2: Divide the secret message into several blocks. Each block contains 2 bits.

Step 3: Store the number of blocks into an array.

Step 4: Initialize a counter: C = 1

Step 5: Find the next embedding position.

- IF all positions are used, then go to step 8.
- IF a line is finished, C = 1

Step 6: Find next block to be embedded.

- IF no block. Go to step 8.
- For i = 1 to no. of blocks
  - IF block[i] = '00'
    go to end of line, then shrink. C = C+1
  - Else IF block[i] = '11'

    go to end of line, then expand. C = C+1
  - Else IF block[i] = '01'
    Embed "0", use two spaces instead of single space in between two
    words.
  - Else IF block[i] = '10'

Embed “1”, use an extra space before any special character.
if there is no special character in a line, we add one special character to
embed ‘1’.
Step 7: End Loop
Step 8: End Algo
Advantages:
Text hiding capacity increases.
** This algorithm is best suited in centrally aligned messages, better in left & right justified messages & worst in justified.

4.2 Proposed Algorithm for Decoding

Step 1: Store the Stego text in an array.
Step 2: Scan the Stego text using ORC software.
Step 3: Find all spaces present in the Stego text.
Step 4: IF 2 spaces exist simultaneously in the text,
then Extract ‘01’
Step 5: Find all special characters successing a space in the Stego text.
Step 6: IF a special character is present after a space in the text,
then Extract ‘10’
Step 7: IF line size < standard line size
then Extract ‘00’
Step 8: IF line size > standard line size
then Extract ‘11’
Step 9: Combine all the extracted bits into an array.
Step 10: End

5. EXPERIMENTAL DETAILS

In this project, files and information were hidden in text documents by the use of the described algorithm. For this purpose, a single cover file is selected. The secret text is compressed and then hidden in the cover text by our steganography algorithm. The steganography algorithm is developed by Matlab and Javascript in Windows Vista platform. In this project, the chosen cover file size and secret file size is 4kb and 1kb respectively. First the secret text is converted to its ASCII value and then is converted to binary bits. Then, the bits are stored in an array which is further divided into blocks of 2 bits each for compressing the secret text. Then, the blocks of bits are embedded in the cover file according to the algorithm and the stego file is generated. For extracting the data from the stego text, it is scanned through OCR software. Then, by our proposed decoding algorithm the bits are extracted.

5.1 Pseudo Code for Encryption Process

5.1.1 Pseudo code for main function:

```
Input: secret text, cover text
Output: Stego text

s= secret text
FOR i:= 1 to length(s)
      Calculate ascii value of s
      Calculate binary value of s
END FOR
     b= array of binary bits of s
 FOR j:= 1 to length(b)
       Split b into blocks of 2 bits each
END FOR
ar= stores the blocked bits array
FOR k:= 1 to length(ar)
     IF ( ar(k) = ‘00’)
        Call Shrink( );
     ELSEIF( ar(k) = ‘11’)
        Call Expand( );
     ELSEIF ( ar(k) = ‘01’)
        Call Addspace( );
     ELSEIF ( ar(k) = ‘10’)
        Call Addspecial( );
     END IF
  END FOR
```

5.1.2 Pseudo code for Expand( ) function:

```
Input:  cover text
Output: Stego text

S2=cover text
C=1
WHILE ( i< length(S2))
    IF( S2(i) != ‘/n’)
      Increase the font size of the line
      i=i+1
     C=C+1
        END IF
    END WHILE
```

5.1.3 Pseudo code for Shrink( ) function:

```
Input:  cover text
Output: Stego text

S2=cover text
C=1
WHILE ( i< length(S2))
    IF( S2(i) != ‘/n’)
      Decrease the font size of the line
      i=i+1
     C=C+1
        END IF
    END WHILE
```

5.1.4 Pseudo code for Addspace( ) function:

```
Input:  cover text
Output: Stego text

S2=cover text
C=1
  a=line for embedding
  s=length(a)
 IF(s == 0)
    C=C+1
ELSE
      Find the embedding position
     Add an extra space in that position
```

```
END IF
    C=C+1
```

5.1.5 *Pseudo code for Addspecial( ) function:*

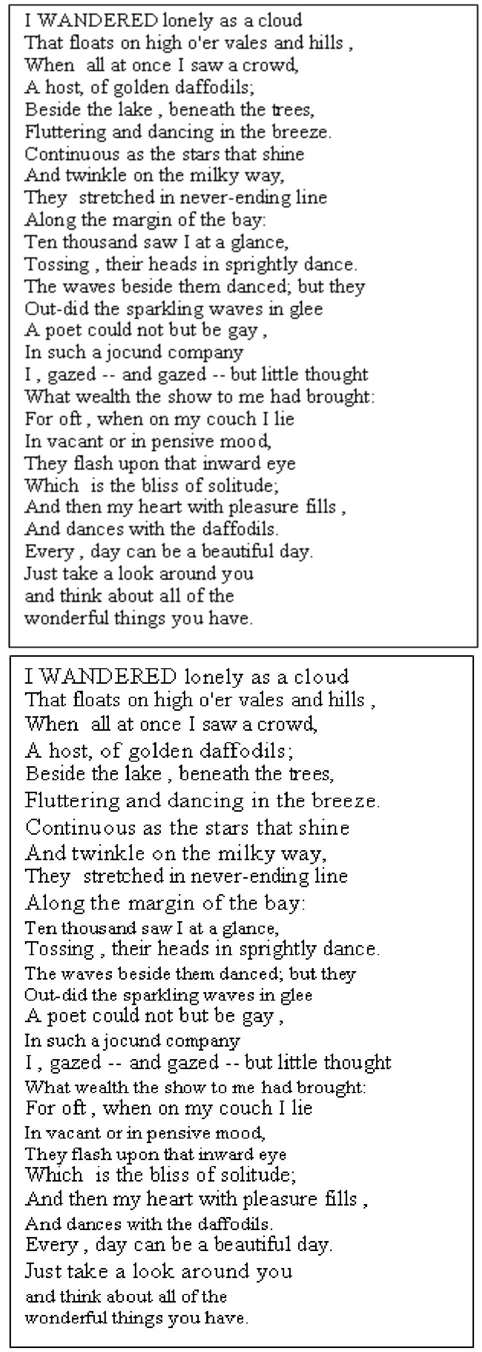
```
Input:  cover text
Output: Stego text

S2=cover text
C=1
  a=line for embedding
  s=length(a)
 IF(s == 0)
    C=C+1
ELSE
      Find the special character position
      Add an extra space in that position
END IF
    C=C+1
```

6. AN EXAMPLE OF THE PROPOSED ALGORITHM

1. The cover file “cover.txt” of size 4KB is taken.
2. The secret file “sss.txt” of size 1KB is taken.
3. First the secret file “sss.txt” is imported and the secret text is stored in an array.
   e.g., a = ‘son D 10’
4. The secret text is converted to its ascii value.
   e.g., sd = 115 111 110 32 68 32 49 48
5. Then the secret texts binary value is calculated.
   e.g., sb = 1110011
   1101111
   1101110
   0100000
   1000100
   0100000
   0110001
   0110000
6. Then the texts binary values are concatenated and stored in an array.
   e.g.,
   SA=11100111101111111011100100000100010001000000110001 0110000
7. Size of the array SA is calculated.
   e.g., no =56 bits
8. Then array SA is divided into blocks of 2 bits each.
   No. Of Blocks = 28
   e.g., k =11, 10, 01, 11, 10, 11, 11, 11, 01, 11, 00, 10, 00, 00, 10, 00, 10,
   00, 10, 00, 00, 01, 10, 00, 10, 11, 00, 00
   8.1 There are 4 block options i.e., 00 , 11 , 01 , 10
   8.2 Block ‘00’ calls Shrink( ) to shrink the line.
   Block ‘11’ calls Expand( ) to expand the line.

Block ‘01’ calls Addspace( ) to add an extra space in between 2 words.
Block ‘10’ calls Addspecial( ) to add an space before a special character.



I WANDERED lonely as a cloud
That floats on high o'er vales and hills ,
When all at once I saw a crowd,
A host, of golden daffodils;
Beside the lake , beneath the trees,
Fluttering and dancing in the breeze.
Continuous as the stars that shine
And twinkle on the milky way,
They stretched in never-ending line
Along the margin of the bay:
Ten thousand saw I at a glance,
Tossing , their heads in sprightly dance.
The waves beside them danced; but they
Out-did the sparkling waves in glee
A poet could not but be gay ,
In such a jocund company
I , gazed -- and gazed -- but little thought
What wealth the show to me had brought:
For oft , when on my couch I lie
In vacant or in pensive mood,
They flash upon that inward eye
Which is the bliss of solitude;
And then my heart with pleasure fills ,
And dances with the daffodils.
Every , day can be a beautiful day.
Just take a look around you
and think about all of the
wonderful things you have.

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Figure 6. Original Cover Text (upper) and Generated Stego Text (down)

7. PERFORMANCE ANALYSIS

According to some existing algorithm based on Inter-paragraph spacing (Inter-sentence) and Inter-word spacing, method is used to encode a binary message into a text by placing either one or two spaces at the end of each terminator character (embedding position). It encodes a ‘0’ by adding a single space and encodes ‘1’ by adding two spaces. The only disadvantage of the existing algorithm is inefficiency because it requires a large amount of text to encode a very few bits. One bit per sentence equates to a data rate of approximately 1 bit per 160 bytes assuming sentences are on average to 80-character lines of text. This method fully depends on a structure of the text. In the embedding method of the existing algorithm, two spaces encode one bit per line, four encode two, eight encode three, etc., gradually increasing the amount of information we want to encode. In right justification method data are encoded by controlling where the extra spaces are placed. One space between words is interpreted as a '0'; two spaces are interpreted as '1'. These methods results in several bits encoded in a line. Because of constraints upon justification not every inter-word space can be used as data. Bender [12] embedded a Manchester like encoding method to determine which of the inter-word spaces represent hidden data bits and which is part of the original text. According to him '01' is interpreted as '1' and '10' is interpreted as '0'. The bit string '00' and '11' are null. According to this method we can embed '01' and '10' at the desired embedding position using some inter-word spacing rule. But when we get '00' 0r '11' as a secret message they represent null string. According to the proposed algorithm we first divide the secret message in two bit format and encode it in a cover file based on some format change of that cover file. Here we are not considering '00' or '11' as null string, rather in these four cases (i.e., 00, 11, 01, 10) we used different inter-word spacing and inter-sentence spacing mechanisms. For example, when the secret message code is '00' the inter-line space is increased i.e., we are expanding the distance between two consecutive lines and when the code is '11' the inter-line space is decreased. When the code is '10' we add a space before a special character and when the code is '01' we add an extra space in between two words. In comparison to existing algorithm the proposed algorithm is robust but high in capacity. According to proposed algorithm with the same cover text we can embed the secret message whose size is twice to the embedded secret message using the existing algorithm. For example, a character is equivalent of 8 bits and it requires approximately eight inter-word spaces to encode one character using the existing algorithm. If we use the proposed algorithm it requires maximum of four inter spaces. Again if there is a combination of '00' and '11' bits then the requirement of the inter space will be low. The time complexity of the existing algorithm is O(n) whereas the time complexity of the proposed algorithm is $O(n^2)$.

8. CONCLUSION AND FUTURE WORK

Text Steganography illustrates a bright prospect because of its widely transmission on today's Internet. In this paper, we introduce a new approach for steganography of information in English texts by embedding the binary values of character texts. Using existing algorithm we can embed the secret message bit-wise. Using proposed algorithm we are embedding a single bit which indicates the value of a secret message block containing two bits. So, we can conclude that, by using the proposed algorithm we can increase the storage capacity of a fixed length cover object compared to the existing algorithm. In addition to establishing secret communication, this method can be used for preventing illegal duplication and distribution of texts especially electronic texts. In addition to use this method for electronic texts, it can be applied on hard copy documents. To this end, we print the document after hiding data in it. For extracting data from the hard copy document, we scan it and unhide the embedded data by computer. The future work should be focused towards optimizing the robustness of the decoding algorithm. This is because the hidden data will be destroyed once the spaces are deleted by some word processing software. Beside that to minimize the computational complexity of the proposed encryption algorithm.

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